

Artifact Appendix

AEGIS: Attention-Embedding Gradient Isolation Shield
Triple-Channel Gradient Masking for Privacy-Preserving Federated LLM Fine-Tuning

Paper ID: 3053

1 Artifact Appendix

Badge. We request *Artifacts Reproduced*. The artifact is a scaled-down CPU micro-GPT-2 experiment with fixed local token IDs and default **AegisDefense**. It supports the directional claim that AEGIS suppresses embedding-gradient token-set leakage versus an undefended baseline.

Requirements. Docker (x86-64/arm64) or Python 3.11 on Linux/macOS. Setup 5–20 minutes; kick-the-tires <2 minutes; scaled-reproduction ~5 minutes (up to ~10 on slower hosts).

Folder structure. The submission package layout is:

```
.
|-- README.md                # install and run guide
|-- Dockerfile               # CPU Docker image
|-- pyproject.toml           # Python package metadata
|-- constraints-cpu.txt       # pinned CPU package versions
|-- THIRD_PARTY_NOTICES.md   # vendored DAGER notice
|-- LICENSES/                # third-party licenses
|-- artifact/
|   |-- appendix.tex          # this appendix source
|   '-- appendix.pdf          # this appendix PDF
|-- scripts/
|   |-- install-artifact.sh   # create .venv and install deps
|   |-- kick-the-tires.sh     # quick test + verify
|   |-- run-scaled-reproduction.sh
|   '-- verify-artifact.py    # check result JSON
|-- configs/studies/          # optional research study configs
'-- src/aegis/
    |-- cli.py                # aegis CLI entry
    |-- artifact/             # kick-the-tires / scaled runners
    |-- attacks/              # reconstruction attacks
    |-- defenses/             # AEGIS and baselines
    |-- evaluation/           # metrics
    |-- harness/              # study runner
    |-- models/               # micro-model helpers
    |-- training/             # training loop
    |-- data/ studies/ config/
```

```
'-- third_party/dager/    # vendored DAGER helpers
```

After a successful run, results are written under `artifact-results/` (created at runtime; not shipped in the package).

Claims. **R1:** undefended embed-norm content-token recall above threshold. **R2:** AEGIS suppresses protected recall below threshold; channels/utility remain finite.

Workflow.

```
docker build -t aegis-artifact:ndss27 .
docker create --name aegis-ktt aegis-artifact:ndss27
docker start --attach aegis-ktt
docker cp aegis-ktt:/opt/aegis/artifact-results/kick-the-tires.json .
docker rm aegis-ktt

docker create --name aegis-repro aegis-artifact:ndss27 \
  /opt/aegis/scripts/run-scaled-reproduction.sh
docker start --attach aegis-repro
docker cp aegis-repro:/opt/aegis/artifact-results/scaled-reproduction.json .
docker rm aegis-repro
```

Locally: `scripts/install-artifact.sh`, then `aegis artifact` (or `quick-test / scaled-reproduction`). Expect exit status 0 and a `Verified` line with `device=cpu` and schema version 3. Full detail is in `README.md`.